MANISH RAINA

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PROFILE SUMMARY

Computer Science (AI & ML) graduate with end-to-end experience building and shipping ML-driven products across NLP, computer vision, predictive analytics, and full-stack applications. Experienced in developing production-oriented ML pipelines and translating technical solutions into measurable outcomes, including 91%+ AUC, 96% classification accuracy, and sub-100ms inference latency. Proficient in Python, Scikit-learn, TensorFlow, FastAPI, React, and modern NLP/ML techniques.

CORE COMPETENCIES

- **Languages:** Python, C, C++, JavaScript, SQL, TypeScript
- **AI Agents & Orchestration:** LangChain, LangGraph, Model Context Protocol (MCP), Multi-Agent Systems, Agentic Tool/Skill Design, Function Calling, Prompt Engineering, RAG Pipelines
- **ML & Model Training:** Scikit-learn, TensorFlow, LightGBM, NLP, Model Training & Fine-Tuning, Embedding-based Semantic Search, Model Evaluation (AUC, Cross-Validation, Calibration, IPCW Weighting)
- **Web & Backend / APIs:** FastAPI, React.js, REST APIs, API Design & Integration, WebSockets, JWT Authentication, RBAC, Middleware Validation
- **Data & Infra:** PostgreSQL, MongoDB, Redis, SQLite, Qdrant Vector DB, Docker, Docker-compose, Git
- **Other:** Web Scraping, Power BI, Speech Recognition & Synthesis, OpenCV, dlib, Playwright

EXPERIENCE

Machine Learning & Web Development Intern — EISystems Technologies
June 2024 – Aug 2024

- **Problem solved:** Generic recommenders ignore customer emotion in reviews, so suggestions match poorly. *Source*
- Built an NLP pipeline (VADER, TF-IDF + Logistic Regression, fine-tuned BERT, DistilRoBERTa) classifying review sentiment at **85–90%** accuracy and emotion at **96%** accuracy.Fed both signals into a similarity-based recommender, raising match quality to **90–95%.**

Machine Learning Research Intern — NIT Delhi
July 2025 – Sept 2025

- **Problem solved:** Predicting heart-disease survival is hard because censored patients bias standard models. *Source*
- With a 3-person team, built a LightGBM + IPCW survival model on 897 patient records, achieving **91.89%** cross-validation AUC Beat a Cox Proportional Hazards baseline by **+15.46%** IPCW AUC at the 180-day horizon. (5-fold CV, 75 tuning runs).

PROJECTS

Nexus AI OS v2.0 — Autonomous Multi-Agent Workspace & AI Orchestration Platform
GitHub

- **Problem solved:** Complex multi-step tasks need several tools and agents coordinated safely, not a single chatbot.
- Engineered an orchestration kernel routing tasks across **12** specialized agents via a JSON-based DAG (30+ sub-actions), with a custom MCP client integrating **6** MCP servers and 10 built-in tools, plus a RAG pipeline (Qdrant/TF-IDF) and 9 safety checkpoints.
- Achieved **4/4** integration tests, 6/6 DAG workflows, and 96% task-step success (24/25); full 4-step workflows complete in **3.8–4.2 seconds.**

AI Career Skill Intelligence Platform (Full-Stack + ML + Scraper)
GitHub

- **Problem solved:** Job seekers cannot easily see which skills they lack for the roles they want.
- Architected a full-stack platform (FastAPI, React, PostgreSQL) with REST endpoints, parsing resumes at **98.2%** extraction accuracy and computing readiness scores in **65.3ms** end-to-end.
- Analyzed 32,217 job listings across 3,947 companies; the hybrid semantic skill extractor reached **88.3%** F1 (vs. 80.4% keyword baseline) and raised recall from **68.2% to 92.5%.**

AI Voice Receptionist (Speech-to-Speech AI Assistant39)
GitHub

- **Problem solved:** Small offices need a hands-free assistant that answers voice queries and recognizes who is speaking.
- Built a speech-to-speech assistant achieving **88%** intent recognition accuracy (44/50 test queries), **1.2–1.6s** speech-to-response latency, and offline pyttsx3 TTS in **80–120ms.**
- Integrated real-time face recognition (DeepFace, VGG-Face, OpenCV) achieving **91.7%** accuracy (55/60 live trials) at **~450–650ms** per frame, plus automated Gmail IMAP email briefing.

EDUCATION

B.Tech, Computer Science Engineering (AI & ML) — PIET, Haryana *Aug 2022 – Aug 2026*

Senior Secondary (Class XII) — Kotwal National Institute *Mar 2021 – Mar 2022*

CERTIFICATIONS & ACHIEVEMENTS

- Machine Learning with Python — Internship Certificate, EISystems Technologies
- Machine Learning Research Internship — NIT Delhi (2025)
- Advanced Diploma in Software Technology (12 Months) — Supertech (India) Computer Education